

Ultimate Amiga OS

Technical Reference Manual

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UAOS is a standalone, bare-metal 64-bit operating system for x86_64 platforms. It is derived from the AROS open-source Amiga replacement kernel and features transparent M68k JIT emulation and native GNU tool integration.

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Chapter 1

Introduction

1.1 Project Overview

Ultimate Amiga OS (UAOS) is a standalone bare-metal x86_64 operating system designed to provide a native Amiga-compatible environment on modern hardware without requiring any host operating system. It achieves this through:

- An extended AROS microkernel providing Exec, DOS, and Intuition semantics on 64-bit hardware.
- A transparent M68k JIT emulation sandbox that executes legacy Amiga “Hunk” binaries inside a 4 GB virtual address space.
- An illegal-opcode trap mechanism (ILLEGAL/0x4AFC) that intercepts system calls at the ROM vector level and dispatches them to native 64-bit C implementations.
- An MMU sandbox using 2 MB huge pages with hardware-register windows deliberately left non-present to transparently virtualise custom chipset I/O via #PF exceptions.

1.2 Architectural Constraints

1. **No Linux kernel.** UAOS boots directly into its own x86_64 AROS-derived microkernel. No Linux, no POSIX kernel layer.
2. **Trap-based thunking.** M68k Hunk binaries are loaded into the 4 GB sandbox. System calls are intercepted via ILLEGAL + \$414D magic-token sequences routed to native 64-bit AROS libraries.
3. **MMU hardware virtualisation.** x86_64 page tables map the sandbox with 2 MB huge pages. The Amiga hardware register window 0x00B00000–0x00DFFFFFF is marked NOT PRESENT.
4. **Coexisting filesystem namespace.** dos.library handlers resolve Amiga assigns (SYS:, S:, C:, RAM:) alongside POSIX translation paths.
5. **AROS Kickstart firmware.** The open-source AROS replacement ROM is used as the modular firmware base with vectors patched at build time.

Chapter 2

System Architecture

2.1 Boot Sequence

1. GRUB2 loads the ELF64 kernel via the Multiboot2 protocol.
2. `uaos_kernel_entry.asm` transitions the CPU from 32-bit protected mode into 64-bit long mode, sets up a bootstrap identity-map page table, and calls `uaos_kernel_main()`.
3. The C kernel entry point validates the Multiboot2 magic value, initialises the VGA text-mode console and UART, then calls:
 - `UAOS_MMU_Init()` — installs the sandbox page tables.
 - `UAOS_ROM_RegisterAll()` — populates the ROM module registry.
 - `UAOS_Bridge_Init()` — allocates the 4 GB guest RAM window.

2.2 M68k Thunk Dispatch

The thunk mechanism replaces selected Exec jump-table vectors in the AROS ROM with 8-byte breakout stubs assembled by the `MK_THUNK` macro in `emulation/rom_patches/rom_traps.s`:

Listing 2.1: `MK_THUNK` macro

```
1 MK_THUNK MACRO
2     ILLEGAL                ; $4AFC --- triggers JIT breakout
3     dc.w    \1             ; UAOS magic token ($414D = "AM")
4     dc.w    \2             ; function index (1-based)
5     rts                   ; return to guest after dispatch
6 ENDM
```

When the JIT core decodes `0x4AFC` it invokes `UAOS_Bridge_IllegalOpcode()`. The C dispatcher:

1. Reads the two words immediately following the `ILLEGAL` opcode.
2. Validates the `$414D` signature.
3. Extracts arguments from the M68k register state (`A1`, `D0`, `D1`, ...).
4. Calls the native 64-bit stub.
5. Writes return values back into `D0/A0`.
6. Advances the guest PC by 6 bytes.

2.3 MMU Sandbox

`kernel/exec/mmu_sandbox.c` constructs four-level x86_64 paging tables mapping the entire 4 GB guest address space using 2 MB huge pages. Table 2.1 shows the critical address assignments.

Address Range	Size	Description
0x00000000-0x001FFFFFFF	2 MB	Chip RAM
0x00200000-0x009FFFFFFF	8 MB	Fast RAM (lower)
0x00B00000-0x00DFFFFFFF	3 MB	Custom chip registers (NOT PRESENT)
0x00F80000-0x00FFFFFFF	512 KB	Kickstart ROM mirror
0x01000000-0x7FFFFFFF	2 GB	Extended Fast RAM
0x80000000-0xFFFFFFFF	2 GB	x86_64 kernel space

Table 2.1: UAOS Guest Address Space Map

Pages in the hardware register window have `PAGE_PRESENT` cleared and `PAGE_NO_CACHE` set. Any access faults to this region are caught by the `#PF` handler in `kernel/exec/page_fault_handler.c` and forwarded to the chip emulator.

Chapter 3

Kernel Subsystems

3.1 ROM Module Registry (`rom_modules.c`)

`UAOS_ROM_RegisterAll()` is called once at boot. It populates a static array of `UaosRomModule` descriptors, each holding:

- Library name string (e.g. `"exec.library"`).
- Library version number.
- 32-bit Amiga base address.
- Array of native function pointers (indexed 1-based to match `rom_traps.s` function indices).

3.2 Thunk Handler (`thunk_handler.c`)

The `UAOS_AMIGA_TO_HOST()` macro translates any 32-bit Amiga guest address to the corresponding host linear address:

Listing 3.1: Address translation macro

```
1 #define UAOS_AMIGA_TO_HOST(amiga_addr) \  
2   ((void *)((uintptr_t)(uos_ram_base) + (uint32_t)(amiga_addr)))
```

3.3 Page Fault Handler (`page_fault_handler.c`)

The ISR entry stub `uos_page_fault_isr` (written in inline GAS assembly) saves all 15 GPRs onto the kernel stack, computes pointers to the `InterruptFrame` and `SavedRegs` blocks, and calls the C function `UAOS_PageFaultHandler()`. On return from the C handler, GPRs are restored and `IRETQ` resumes execution past the faulting instruction.

Chapter 4

Build System

4.1 Prerequisites

Listing 4.1: Install build dependencies (Ubuntu/Debian)

```
1 sudo apt install nasm gcc binutils xorriso \  
2     grub-pc-bin grub-common \  
3     binutils-m68k-linux-gnu python3
```

4.2 Building the ISO

Listing 4.2: Full build pipeline

```
1 # Assemble NASM kernel entry  
2 nasm -f elf64 kernel/boot/uaos_kernel_entry.asm \  
3     -o build/uaos_kernel_entry.o  
4  
5 # Compile C kernel sources (freestanding)  
6 gcc -ffreestanding -fno-stack-protector -fno-pie \  
7     -mno-red-zone -nostdlib -m64 -O2 \  
8     -c kernel/boot/uaos_kernel_main.c -o build/uaos_kernel_main.o  
9 # ... (see scripts/build_iso.sh for full compile step list)  
10  
11 # Link ELF64 kernel  
12 ld -T kernel/boot/uaos_kernel.ld build/*.o -o build/uaos-kernel.elf  
13  
14 # Package ISO  
15 ./scripts/build_iso.sh
```

4.3 Testing in QEMU

Copy the OVMF variables file once (required for UEFI variable store):

Listing 4.3: One-time OVMF setup

```
1 cp /usr/share/OVMF/OVMF_VARS_4M.fd /tmp/ovmf_vars.fd
```

Listing 4.4: QEMU test invocation

```
1 qemu-system-x86_64 \  
2     -machine q35,usb=off \  
3
```

```
3 -drive if=pflash,format=raw,readonly=on,\  
4 file=/usr/share/OVMF/OVMF_CODE_4M.fd \  
5 -drive if=pflash,format=raw,file=/tmp/ovmf_vars.fd \  
6 -cdrom build/Ultimate_Amiga_OS.iso \  
7 -m 512M -vga virtio -no-reboot
```

The `usb=off` flag is required to prevent QEMU's USB tablet device from conflicting with the PS/2 relative mouse driver. Add `-serial stdio` to see kernel debug output on the terminal.

Chapter 5

Graphics and Window Manager

5.1 Framebuffer Initialisation

GRUB2 fills the Multiboot2 framebuffer tag (type 8) from the UEFI GOP driver. `FB_Init()` parses this tag to populate the global `FbState g_fb` structure, which holds the physical base address, pixel dimensions, pitch (bytes per row), and bit depth (24 or 32 bpp).

All drawing primitives (`FB_FillRect`, `FB_DrawHLine`, `FB_DrawVLine`, `FB_PutPixel`, `FB_PutStr`) clip coordinates to the framebuffer bounds before writing, supporting windows that extend partially off-screen.

5.2 Desktop

`Desktop_Draw()` renders the full Workbench-style backdrop:

- **Menu bar** (top, 20 px) — Workbench menu labels and clock area.
- **Backdrop** — light grey base with a dark-grey stipple dot overlay, one dot per two pixels alternating rows/columns (classic Amiga pattern).
- **Disk icons** — top-right corner, stacked vertically.
- **Information window** — centred boot status window.
- **Status bar** (bottom, 18 px) — kernel idle message and RAM size.

`Desktop_RedrawRect(rx, ry, rw, rh)` repaints a sub-rectangle of the desktop (backdrop + overlays) without a full-screen repaint. This is used by the window manager to erase the old window footprint during drag and resize, avoiding the full-screen flicker that would otherwise occur.

5.3 Window Manager

`kernel/display/wm.c` implements a lightweight window manager supporting up to eight windows simultaneously.

5.3.1 Data Structures

Listing 5.1: WmWindow structure

```
1 typedef struct {  
2     int     x, y, w, h;
```

```
3     char        title[32];
4     WM_DrawFn   draw;        /* repaint callback: (x, y, w, h) */
5     WM_KeyFn    on_key;      /* keystroke callback: (char c)   */
6     int         active;
7 } WmWindow;
```

5.3.2 Z-Order and Focus

Windows are stored in `g_wins[]` and indexed by a separate `g_zorder[]` array (`[0]` = back, `[g_nwins-1]` = top). `WM_Redraw()` paints windows back-to-front. Clicking a window calls `raise_window()` to promote it to the top of the z-stack.

5.3.3 Drag and Resize

- **Drag** — a mouse-down on the title bar records the grab offset. Each subsequent mouse move erases the old footprint with `Desktop_RedrawRect`, updates `w->x/y`, and repaints all windows.
- **Resize** — a mouse-down on the 16×16 resize grip in the bottom-right corner records the base size and origin. Mouse movement computes the new dimensions relative to the drag start point.
- Windows may extend off the left, right, and bottom screen edges. The title bar is always constrained to remain on-screen.

5.4 Software Cursor

`kernel/display/cursor.c` implements a 16×16 Amiga-style arrow sprite cursor. The cursor is rendered by saving the background pixels beneath it, drawing the sprite, and on each move restoring the saved background before re-saving and re-drawing at the new position. This ensures the desktop and window content are never permanently overwritten by cursor movement.

Chapter 6

Troubleshooting

6.1 GRUB: “Invalid arch-dependent ELF magic”

This error means GRUB received a file that is not a valid ELF64 executable. Ensure `uaos-kernel.elf` was produced by the full NASM + GCC + LD pipeline (not the Python mock generator used for structural validation only).

6.2 GRUB: “You need to load the kernel first”

GRUB could not find the `multiboot2` header within the first 32 KB of the binary. Verify the linker script places `.multiboot2` first and that the NASM source emits the section correctly.

6.3 Kernel halts immediately after boot

Enable serial output and check the UART log:

```
1 qemu-system-x86_64 -cdrom build/Ultimate_Amiga_OS.iso \  
2 -m 512M -boot d -serial stdio 2>&1 | tee boot.log
```

Appendix A

Amiga Memory Map Reference

See Table [2.1](#) in Chapter 2. The full OCS/ECS/AGA custom chip register map follows Commodore's *Hardware Reference Manual, 3rd Ed.*